

28 May / Mei 2012

Surname and initials	<u>Memo</u>	Student number	_____
<i>Van en voorletters</i>	_____	<i>Studentenommer</i>	_____
Signature	_____		
<i>Handtekening</i>			
Time	2 hours	Max. Total	85 Marks
<i>Tyd</i>	<i>2 ure</i>	<i>Maks. Totaal</i>	<i>85 Punte</i>
Internal Examiner:	Dr J M Nel	External Examiner	Mr R Q Odendaal
<i>Interne Eksaminator</i>		<i>Eksterne Eksaminator</i>	

• Answer all questions. • Clearly show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • No textbooks allowed. • A formula sheet is included. • All test and examination regulations of the University of Pretoria are applicable.	• <i>Beantwoord alle vrae.</i> • <i>Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.</i> • <i>Stel duidelik alle aannames, reëls en wette wat u gebruik.</i> • <i>Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).</i> • <i>Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte nie noodwendig vir 'n numeries-korrekte antwoord gegee sal word nie.</i> • <i>Programmeerbare sakrekenaars MAG NIE gebruik word nie.</i> • <i>Geen handboeke word toegelaat nie.</i> • <i>'n Formuleblad word ingesluit.</i> • <i>Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.</i>
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Semester Test (Aegrotat)

FSK 116
28 May / Mei 2012

Semestertoets (Siekte)

Surname and initials	_____	Student number	_____
<i>Van en voorletters</i>		<i>Studentenommer</i>	
Signature	_____	Venue	NW1 5-42
<i>Handtekening</i>		<i>Lokaal</i>	

Question 1

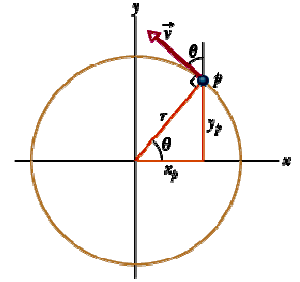
15

(a) Consider a particle P , which is moving at a constant speed v , around a circle path of radius r . At the instant shown P has co-ordinates x_p and y_p . Stating all assumptions, and showing all your steps, show that the magnitude of the

acceleration of particle P is given by $a = \frac{v^2}{r}$.

Beskou 'n deeltjie P , wat met 'n konstante spoed v beweeg in 'n sirkelbaan met radius r . Op die tydstip aangedui in die figuur het P die koördinate x_p en y_p . Wys duidelik al jou stappe en stel duidelik al jou aannames, en wys dat die grootte

van die versnelling van deeltjie P $a = \frac{v^2}{r}$ is.



[6]

coordinates: x_p, y_p

position: $\vec{r}$ is at angle θ to x -axis

$\vec{v} \perp$ position

$\vec{v}$ is at angle θ to perpendicular at P (as shown in Fig)

$$\vec{v} = v_x \hat{i} + v_y \hat{j} = (-v \sin \theta) \hat{i} + (v \cos \theta) \hat{j}$$

$$\vec{v} = \left(-v \left(\frac{y_p}{r} \right) \right) \hat{i} + \left(v \left(\frac{x_p}{r} \right) \right) \hat{j} \quad \vec{v} = \left(-v \left(\frac{y_p}{r} \right) \right) \hat{i} + \left(v \left(\frac{x_p}{r} \right) \right) \hat{j}$$

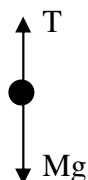
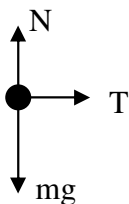
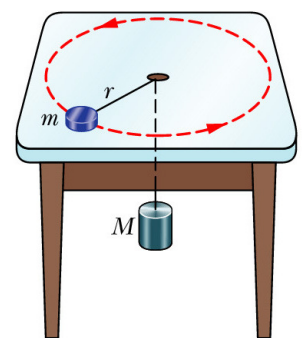
$$\vec{a} = \frac{d\vec{v}}{dt} = \left(-\frac{v}{r} \left(\frac{dy_p}{dt} \right) \right) \hat{i} + \left(\frac{v}{r} \left(\frac{dx_p}{dt} \right) \right) \hat{j} = \left(-\frac{v^2}{r} \cos \theta \right) \hat{i} + \left(\frac{v^2}{r} \sin \theta \right) \hat{j}$$

Magnitude of acceleration:

$$a = \sqrt{a_x^2 + a_y^2} = \frac{v^2}{r} \sqrt{\cos^2 \theta + \sin^2 \theta} = \frac{v^2}{r}$$

(b) A puck of mass $m = 1.50$ kg slides in a circle of radius $r = 20.0$ cm on a frictionless table while attached to a hanging cylinder of mass $M = 2.50$ kg by a cord through a hole in the table. Show that the speed of the puck must be 1.81 m/s for the cylinder to be at rest.

'n Skuif met massa $m = 1.50$ kg beweeg in 'n sirkel met radius $r = 20.0$ cm op 'n wrywinglose tafel, terwyl 'n silinder met massa $M = 2.50$ kg van 'n koord hang wat deur 'n gaatjie in die lê. Wys dat die spoed van die skuif 1.81 m/s moet wees om te die silinder te laat stilstaan.



[2+5]

For the puck to have uniform circular motion and cylinder to remain at rest

$$\sum F_{m,x} = ma_x = T, \text{ where } a = \frac{v^2}{r}$$

$$\sum F_{M,y} = Ma_y = 0$$

$$0 = T - Mg$$

Therefore $T = Mg$

And from first equation $T = mv^2/r$.

$$Mg = mv^2/r.$$

$$v = \sqrt{\frac{Mgr}{m}} = \sqrt{\frac{(2.50 \text{ kg})(9.8 \text{ m/s}^2)(0.200 \text{ m})}{1.50 \text{ kg}}} = 1.81 \text{ m/s}$$

(c) Determine the period of the puck.

Bepaal die periode van die skuif.

[2]

$$T = \frac{2\pi r}{v} = 0.69 \text{ s}$$

(d) Determine the angular velocity of the puck.

Bepaal die hoeksnelheid van die skuif.

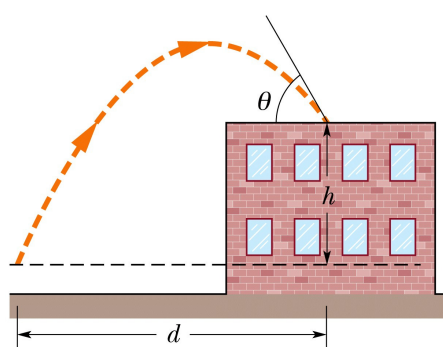
[2]

$$\omega = \frac{v}{r} = \frac{1.81}{0.2} = 9.05 \text{ rad/s}$$

$$\text{or } T = 2\pi/\omega$$

Question 2

10



A ball is thrown up onto a roof, landing 4.50 s later at a height $h = 20 \text{ m}$ above the release level; refer to the figure. The ball's path just before landing is angled at $\theta = 60^\circ$ with the roof.

'n Bal word tot op 'n dak gegooi, en land 4.50 s later op 'n hoogte van $h = 20 \text{ m}$ bo die lanseervla; verwys na die figuur. Die bal se pad net voor dit land maak 'n hoek van $\theta = 60^\circ$ met die dak.

(a) Determine the horizontal distance d the ball travels.

Bepaal die horisontale afstand d wat die bal aflê.

[5]

$$\begin{aligned} t &= 4.50 \text{ s} \\ a_y &= -9.8 \text{ m/s}^2 \\ y &= 20 \text{ m} \\ v_{oy} &= ? \\ v_y &= ? \text{ at } 4.5 \text{ s} \end{aligned}$$

$$\begin{aligned} a_x &= 0 \text{ m/s}^2 \\ x &= ? \\ v_{ox} &= ? \end{aligned}$$

$$\Delta y = v_{y_i} t - \frac{1}{2} a t^2$$

$$v_{y_i} = \frac{\Delta y + \frac{1}{2} a t^2}{t} = \frac{+20 \text{ m} + \frac{1}{2} (-9.8 \text{ m/s}^2) (4.50 \text{ s})^2}{4.50 \text{ s}}$$

$$= -17.60 \text{ m/s}$$

$$\tan \theta = \frac{v_{y_i}}{v_{x_i}}$$

$$v_{x_i} = \frac{v_{y_i}}{\tan \theta} = \frac{17.6}{\tan 60}$$

$$= 10.16 \text{ m/s}$$

Horizontal distance:

$$\Delta x = v_{x_i} t - \frac{1}{2} a_x t^2$$

$$= 10.16 \text{ m/s} (4.5 \text{ s}) = 45.72 \text{ m}$$

(b) What is the magnitude and direction (relative to the horizontal) of the ball's initial velocity?

Wat is die grootte en rigting (relatief tot die horisontale as) van die bal se aanvanklike snelheid?

[5]

$$\Delta y = v_{oy_i} t + \frac{1}{2} a t^2$$

$$v_{oy_i} = \frac{\Delta y - \frac{1}{2} a t^2}{t} = \frac{+20 \text{ m} - \frac{1}{2} (-9.8 \text{ m/s}^2) (4.50 \text{ s})^2}{4.50 \text{ s}}$$

$$= 26.50 \text{ m/s}$$

No acceleration in x-direction therefore $v_{ox_i} = v_{x_i} = 10.16 \text{ m/s}$

Magnitude of initial velocity: $v_o = \sqrt{v_{ox_i}^2 + v_{oy_i}^2} = 28.4 \text{ m/s}$

Direction:

$$\tan \theta = \frac{v_{oy_i}}{v_{ox_i}}$$

$$\theta = 68.9^\circ \text{ above the horizontal}$$

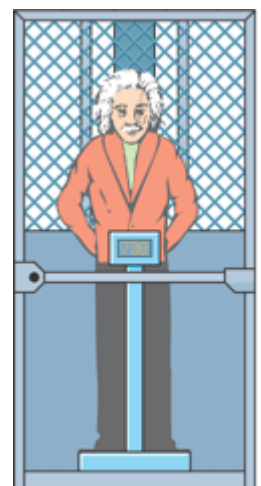
Question 3

In the figure, a passenger of mass $m = 72.2 \text{ kg}$ stands on a platform scale in an elevator cab. We are concerned with the scale readings when the cab is stationary and when it is moving up or down.

Die figuur dui 'n passasier, met massa $m = 72.2 \text{ kg}$, aan wat in 'n hysbak op 'n platform-skaal staan. Ons stel belang in die lesing op die skaal wanneer die hysbak stilstaan en wanneer dit op en af beweeg.

(a) Draw the free body diagram and find a general solution for the scale reading, whatever the vertical motion of the elevator cab.

Teken die vry-liggaams diagram en bepaal die algemene oplossing vir die lesing op die skaal, ongeag die vertikale beweging van die hysbak.



(a)

11

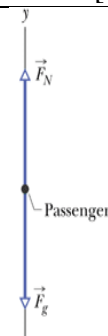
[2+2]

Because the two forces on the passenger and his acceleration are all directed vertically, along the y axis, we can use Newton's second law written for y components

$$\vec{F}_{net} = m\vec{a}$$

$$F_{net,y} = F_N - mg = ma$$

$$F_N = ma + mg = m(a + g)$$



(b) What does the scale read if the cab accelerates upward at 3.20 m/s^2 ?

Wat is die lesing op die skaal wanneer die hysbak opwaarts versnel teen 3.20 m/s^2 ?

[3]

$$a = 3.2 \text{ ms}^{-2}$$

$$F_N = m(a + g)$$

$$= 72.2(3.2 + 9.8)$$

$$= 939 \text{ N}$$

(c) During the upward acceleration in part (b), what is the magnitude F_{net} of the net force on the passenger, **AND** what is the magnitude of the passengers relative acceleration $a_{p,cab}$ as measured in the frame of the cab?

*Tydens die opwaartse versnelling in deel (b), wat is die grootte F_{net} , die netto krag, op die passasier, **EN** wat is die grootte van die passasier se relatiewe versnelling, $a_{p,cab}$, soos gemeet in die raamwerk van die hysbak?*

[4]

$$\vec{F}_{net} = m\vec{a} = 72.2(3.2) = 231 \text{ N}$$

or

$$F_{net,y} = F_N - mg = 939 - (72.2(9.8)) = 231 \text{ N}$$

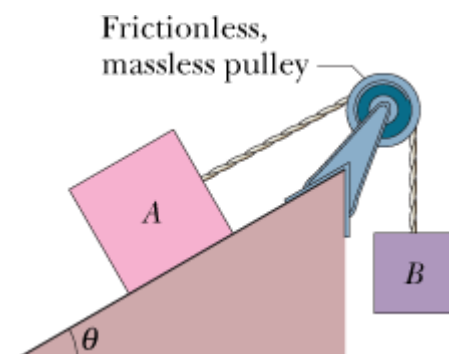
$$a_{p,cab} = 0 \text{ m/s}^2$$

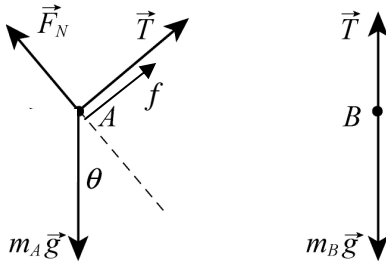
Question 4

7

In the figure, two blocks are connected over a massless pulley. The mass of block A is 10 kg , and the coefficient of kinetic friction between A and the incline is $\mu_k = 0.20$. Angle θ of the incline is 30° . Block A slides down the incline at constant speed. What is the mass of block B?

Die figuur dui twee blokke wat aanmekaar gekonnekteer word oor 'n massalose katrol. Die massa van blok A is 10 kg en die wrywingskoëffisiënt tussen blok A en die skuinste is $\mu_k = 0.20$. Die hoek θ is 30° . Blok A gly by die skuinste af teen 'n konstante spoed. Wat is die massa van blok B?





[2+5]

The free-body diagrams are shown. T is the magnitude of the tension force of the string, f is the magnitude of the force of friction on block A, F_N is the magnitude of the normal force of the plane on block A, $m_A \vec{g}$ is the force of gravity on body A (where $m_A = 10 \text{ kg}$), and $m_B \vec{g}$ is the force of gravity on block B. $\theta = 30^\circ$ is the angle of incline. For A we take the $+x$ to be uphill and $+y$ to be in the direction of the normal force; the positive direction is chosen *downward* for block B.

Since A is moving down the incline, the force of friction is uphill with magnitude $f_k = \mu_k F_N$ (where $\mu_k = 0.20$). Newton's second law leads to

$$\begin{aligned}\sum F_{A,x} &= T + f_k - m_A g \sin \theta = m_A a = 0 \\ \sum F_{A,y} &= F_N - m_A g \cos \theta = 0 \\ \sum F_{B,y} &= m_B g - T = m_B a = 0\end{aligned}$$

for the two bodies (where $a = 0$ is a consequence of the velocity being constant). We solve these for the mass of block B.

$$\begin{aligned}F_N &= m_A g \cos \theta \\ T &= -f_k + m_A g \sin \theta \\ &= -\mu_k F_N + m_A g \sin \theta\end{aligned}$$

Then substituting T into equation for block B in y-direction:

$$\begin{aligned}m_B &= \frac{-\mu_k F_N + m_A g \sin \theta}{g} \\ &= \frac{-\mu_k m_A g \cos \theta + m_A g \sin \theta}{g} \\ m_B &= m_A (\sin \theta - \mu_k \cos \theta) = 3.3 \text{ kg}.\end{aligned}$$

Question 5

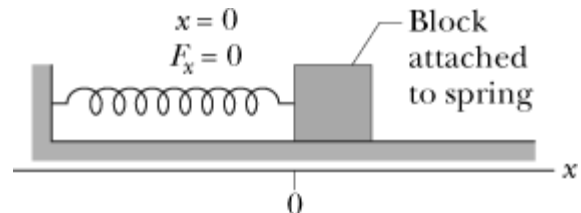
7

Derive the general expression to determine the work done by a massless, ideal spring as it moves a block of mass m , through a distance d . Start from the basic definition of work - $W = \vec{F} \cdot \vec{d} = Fd \cos \phi$.

Assume that the block is on a frictionless surface.

Herlei die algemene vergelyking vir die bepaling van die arbeid verrig deur 'n massalose, ideale

veer, soos dit 'n blok met massa m , deur 'n afstand d beweeg. Begin met die basiese definisie vir arbeid - $W = \vec{F} \cdot \vec{d} = Fd \cos \phi$. Veronderstel dat die blok op 'n wrywinglose oppervlak beweeg.



[7]

For a spring $\vec{F} = -k\vec{d}$ which is a variable force.

Cannot apply above formula for work directly – but divide the distance into small increments Δx over which F can be considered a constant. *Important to state why we cannot use definition given.*

The sum of the work done over all the Δx will then give the total work, $W = \sum F \Delta x$.

In the limit $\Delta x \rightarrow 0$ we can write:

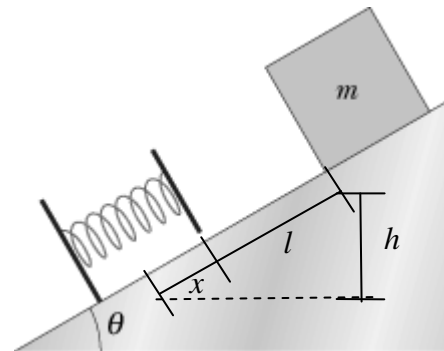
$$\begin{aligned}
 W &= \int_{x_i}^{x_f} F(x) dx \\
 &= \int_{x_i}^{x_f} (-kx) dx = -k \int_{x_i}^{x_f} (x) dx \\
 &= -k \left[\frac{1}{2} x^2 \right]_{x_i}^{x_f} \\
 W &= \frac{1}{2} k x_{x_i}^2 - \frac{1}{2} k x_{x_f}^2
 \end{aligned}$$

Question 6

12

In the figure, a block of mass $m = 12 \text{ kg}$ is released from rest on a frictionless incline of angle $\theta = 30^\circ$. Below the block is a spring that can be compressed 2.0 cm by a force of 270 N. The block momentarily stops when it compresses the spring by 5.5 cm.

Die figuur dui 'n blok van massa $m = 12 \text{ kg}$ aan. Die blok word van rus gelos op 'n wrywinglose skuinsvlak met 'n hoek $\theta = 30^\circ$. Onder op die skuinsvlak is daar 'n veer wat 2.0 cm saamgepers kan word deur 'n krag van 270 N. Die blok stop vir 'n oomblik nadat dit die veer 5.5 cm saamgepers het.



(a) How far does the block move down the incline from its rest position to this stopping point?
Hoe ver beweeg die blok af lanks die skuinsvlak voordat dit tot rus kom van die oorspronklike posisie?

[8]

We refer to its starting point as A , the point where it first comes into contact with the spring as B , and the point where the spring is compressed $|x| = 0.055 \text{ m}$ as C . Point C is our reference point for computing gravitational potential energy. Elastic potential energy (of the spring) is zero when the spring is relaxed. Information given in the second sentence allows us to compute the spring constant. From Hooke's law, we find

$$k = \frac{F}{x} = \frac{270 \text{ N}}{0.02 \text{ m}} = 1.35 \times 10^4 \text{ N/m} . \quad \checkmark \checkmark$$

(a) The distance between points A and B is $\vec{F}_g$ and we note that the total sliding distance $\ell + |x|$ is related to the initial height h of the block (measured relative to C) by

$$\frac{h}{\ell + |x|} = \sin \theta \quad \checkmark$$

where the incline angle θ is 30° . Mechanical energy conservation leads to

$$K_A + U_A = K_C + U_C$$

$$0 + mgh = 0 + \frac{1}{2} kx^2 \quad \checkmark \checkmark$$

which yields

$$h = \frac{kx^2}{2mg} = \frac{(1.35 \times 10^4 \text{ N/m})(0.055 \text{ m})^2}{2(12 \text{ kg})(9.8 \text{ m/s}^2)} = 0.174 \text{ m} . \quad \checkmark \checkmark$$

Therefore,

$$\ell + |x| = \frac{h}{\sin 30^\circ} = \frac{0.174 \text{ m}}{\sin 30^\circ} = 0.35 \text{ m} . \quad \checkmark$$

(b) What is the speed of the block just as it touches the spring?

Wat is die spoed van die blok as dit die veer tref?

[4]

From this result, we find $l' = 0.35 - 0.055 = 0.29$, which means that $\Delta y = -l' \sin \theta = -0.15 \text{ m}$

$\Delta y = -\ell \sin \theta = -0.15 \text{ m}$ in sliding from point A to point B . Thus, Eq. 8-18 gives

$$\Delta K + \Delta U = 0$$

$$\frac{1}{2}mv^2 + mg\Delta y = 0 \quad \checkmark \checkmark$$

which yields $v_B = \sqrt{-2g\Delta y} = \sqrt{-2(9.8 \text{ m.s}^{-2})(-0.15 \text{ m})} = 1.7 \text{ m.s}^{-1}$

Question 7**12**

A soccer player kicks a soccer ball of mass 0.45 kg that is initially at rest. The player's foot is in contact with the ball for 3.0×10^{-3} s, and the force of the kick is given by

$$F(t) = \left[(6.0 \times 10^6)t - (2.0 \times 10^9)t^2 \right] \text{ N}$$

for $0 \leq t \leq 3.0 \times 10^{-3}$ s, where t is in seconds. Calculate the magnitudes of the following:

'n Sokkerspeler skop ' sokkerbal, met massa 0.45 kg wat aanvanklik met rus gelê het. Die speler se voet is in kontak met die bal vir 3.0×10^{-3} s, en die krag van die skop word gegee deur

$$F(t) = \left[(6.0 \times 10^6)t - (2.0 \times 10^9)t^2 \right] \text{ N}$$

vir $0 \leq t \leq 3.0 \times 10^{-3}$ s, waar t in sekondes is. Bereken die groottes van die volgende:

(a) The impulse on the ball due to the kick.

Die impuls op die bal as gevolg van die skop.

[5]

We take the force to be in the positive direction, at least for earlier times. Then the impulse is

$$\begin{aligned} J &= \int_0^{3.0 \times 10^{-3}} F dt = \int_0^{3.0 \times 10^{-3}} \left[(6.0 \times 10^6)t - (2.0 \times 10^9)t^2 \right] dt \\ &= \left[\frac{1}{2} (6.0 \times 10^6)t^2 - \frac{1}{3} (2.0 \times 10^9)t^3 \right] \bigg|_0^{3.0 \times 10^{-3}} \\ &= 9.0 \text{ N} \cdot \text{s}. \end{aligned}$$

(b) The average force on the ball from the player's foot during the period of contact.

Die gemiddelde krag op die bal van die speler se voet gedurende die tydperk van kontak.

[3]

Since $J = F_{\text{avg}} \Delta t$, we find

$$F_{\text{avg}} \frac{J}{\Delta t} = \frac{9.0 \text{ N} \cdot \text{s}}{3.0 \times 10^{-3} \text{ s}} = 3.0 \times 10^3 \text{ N}.$$

(c) The maximum force on the ball from the player's foot during the period of contact.

die maksimum krag op die bal van die speler se voet gedurende die tydperk van kontak.

[4]

To find the time at which the maximum force occurs, we set the derivative of F with respect to time equal to zero – and solve for t .

$$\begin{aligned} \frac{dF}{dt} &= \frac{d}{dt} \left((6.0 \times 10^6)t - (2.0 \times 10^9)t^2 \right) \\ 0 &= (6.0 \times 10^6) - 2(2.0 \times 10^9)t \end{aligned}$$

Then $t = 1.5 \times 10^{-3}$ s.

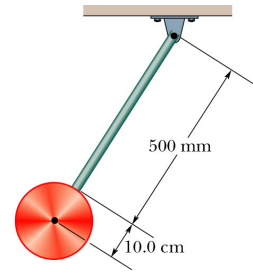
This is the time at which F is a max., so

$$F_{\text{max}} = (6.0 \times 10^6)(1.5 \times 10^{-3}) - (2.0 \times 10^9)(1.5 \times 10^{-3})^2 = 4.5 \times 10^3 \text{ N}.$$

Question 8

The pendulum in the figure consists of a uniform disk with radius 10.0 cm and mass 500 g attached to a uniform thin rod with length 500 mm and mass 270 g.

Die slinger in die figuur bestaan uit 'n univorme skyf met radius van 10.0 cm en massa 500 g wat vasgemaak is aan 'n univorme dun balk van lengte 500 mm en massa 270 g.



(a) Show clearly (all in-between steps) that the rotational inertia of the pendulum about the pivot point is $0.205 \text{ kg}\cdot\text{m}^2$.

Dui duidelik aan (alle tussenstappe) dat die traagheidsmoment van die slinger om die draaipunt $0.205 \text{ kg}\cdot\text{m}^2$ is.

[5]

For the disk:

$$\begin{aligned} I_{\text{com,disk}} &= \frac{1}{2}MR^2 \\ I_{\text{disk}} &= \frac{1}{2}MR^2 + Mh^2 = \frac{1}{2}MR^2 + M(R+L)^2 \\ &= \frac{1}{2}(0.500\text{kg})(0.10\text{m})^2 + (0.500\text{kg})(0.10\text{m} + 0.500\text{m})^2 \\ &= 0.1825\text{kg}\cdot\text{m}^2 \end{aligned}$$

For the rod:

$$\begin{aligned} I_{\text{rod}} &= I_{\text{com,rod}} + Mh^2 = \frac{1}{12}ML^2 + Mh^2 \\ &= \frac{1}{12}(0.270\text{kg})(0.5\text{m})^2 + (0.270)(0.25)^2 \\ &= 0.0225\text{kg}\cdot\text{m}^2 \end{aligned}$$

$$\begin{aligned} I_{\text{pendulum}} &= I_{\text{disk}} + I_{\text{rod}} = 0.1825 + 0.0225 \\ &= 0.205\text{kg}\cdot\text{m}^2 \end{aligned}$$

(b) What is the distance between the pivot point and the center of mass of the pendulum?

Wat is die afstand tussen die draaipunt en die massamiddelpunt van die slinger?

[4]

$$x_{\text{cm}} = \frac{1}{M} \sum m_i r_i = \frac{1}{m_{\text{rod}} + m_{\text{disk}}} [m_{\text{rod}} r_{\text{rod}} + m_{\text{disk}} r_{\text{disk}}]$$

$$\begin{aligned} \text{Centre of mass: } &= \frac{1}{0.270 + 0.500} [0.270(0.25) + (0.500)(0.10 + 0.50)] \\ &= 0.4772\text{m} = 47.7\text{cm} \end{aligned}$$

(c) Calculate the period of oscillation.

Bereken die periode van die ossilasie.

[3]

$$\begin{aligned} T &= 2\pi \sqrt{\frac{I}{mgh}} = 2\pi \sqrt{\frac{0.205}{0.770(9.8)(0.477)}} \\ &= 1.50\text{s} \end{aligned}$$

(d) What is the length of a simple pendulum that will have the same period as the physical pendulum in this setup?

Wat is die lengte van die eenvoudige slinger wat die selfde periode sal hê as die fisiese slinger in hierdie opstelling?

[3]

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$L = \left(\frac{T}{2\pi}\right)^2 g$$

$$= \left(\frac{1.5}{2\pi}\right)^2 9.8 = 0.56 \text{ m}$$

Finish!

Klaar!